

Filling the Gaia DR4 Astrometry Gap: Enabling future direct imaging observations with the Lazuli Observatory using high-precision HST Fine Guidance Sensor Astrometry

Abstract

The upcoming Lazuli Observatory, set to launch around 2030, will directly image Jovian exoplanets at optical wavelengths. To design efficient direct imaging campaigns with the Lazuli coronagraph, we must know when and where in their orbits Jovian planets can be detected. However, we currently cannot confidently design direct imaging campaigns for the Lazuli high-priority exoplanet targets; though decades of radial velocity monitoring campaigns have been performed for these exoplanets, current and soon-to-be available absolute astrometry is insufficient to fully constrain these planets' orbits. We propose an astrometric observation program with the HST Fine Guidance Sensors (FGS) to mitigate this astrometry deficiency for the Lazuli target planets. We will obtain multi-epoch HST/FGS astrometry at the sub-milliarcsecond precision required to constrain the orbits of Lazuli's high-priority exoplanets, enabling efficient observation campaign design. This program will be instrumental in enabling reflected light detections only possible with the large time allocation devoted to the Lazuli coronagraph. Beyond their use in designing direct imaging campaigns, the dynamical mass and position constraints arising from this program will be critical to the scientific interpretation of Lazuli's direct imaging observations, particularly in comparing photometric measurements of Jovian planets to atmospheric models.

- **Scientific Justification**

Precise orbital constraints are essential to exoplanet direct imaging with Lazuli.

Designing targeted exoplanet direct imaging campaigns for a coronagraph instrument like the Lazuli ExtraSolar Coronagraph (ESC) requires precise knowledge of the planet’s mass and position relative to its host star. Looking to the Nancy Grace Roman Space Telescope Coronagraph Instrument (CGI) for guidance, CGI’s observation designs demonstrate strong dependence on this prior knowledge. In designing observations for ϵ Eridani (Eri) b, the CGI team is first considering the relative separation of ϵ Eri b from its host star over time to establish time windows when ϵ Eri b is present within the region of high contrast of the instrument where planets can be imaged (the “dark hole”) (S. G. Wolff, 2026). Furthermore, to calculate exposure times for observations, the mass of the planet must be estimated to inform models leading to a planet-to-star flux ratio used in exposure time calculation. Direct imaging observations, regardless of observatory, depend on prior knowledge of exoplanet position and mass to inform the scheduling and exposure time of observations seeking to detect these planets.

However, planet position and mass prior knowledge comes from radial velocity and absolute astrometric observations of at least one full orbit of a given exoplanet, but many of the radial velocity-detected Jovian planets that will be observed by Lazuli ESC lack one full orbit of absolute astrometry. This is a particular problem for Lazuli ESC, as the Lazuli dark hole is a semicircle off to one side of the host star (A. Roy et al., 2026). To design observations with this semicircular dark hole, the two-dimensional location (separation and position angle, not only separation) of the targeted exoplanet relative to its host star must be confidently constrained. While there is an abundance of literature radial velocities available for the Lazuli exoplanet targets over the last 35 years (C. K. Harada et al., 2025), the 66-month time span that will be covered in Gaia DR4 is insufficient to cover one complete orbit of many of these exoplanets. Gaia DR5 will have data from the full 12-year Gaia mission that would solve this problem, but it will not be released until after the likely start date of Lazuli observations (ESA 2026, A. Roy et al. 2026). It is therefore imperative that we begin filling this astrometric information gap now before Lazuli launches.

We propose HST Fine Guidance Sensor (HST/FGS) astrometric observations of selected high-priority Lazuli ESC exoplanet targets without a complete orbit covered by Gaia DR4. HST/FGS will provide the sub-milliarcsecond precision absolute astrometry needed to complete the orbital coverage of these targets. Each target will be visited multiple times, with each visit consisting of two orbits of HST/FGS observations. Each visit will produce an ensemble of independent astrometric measurements that will be combined to produce a single sub-milliarcsecond precision point for each visit. This multi-visit astrometry will be used to constrain the orbits of the Lazuli ESC exoplanet targets to design direct imaging campaigns. The absolute astrometry catalog resulting from this program will further go on to inform the scientific interpretation of the direct imaging observations taken by Lazuli ESC by allowing, for example, comparison of ESC photometric measurements to reflected light atmospheric models that depend on an estimate of the planet’s mass.

Gaia DR4 astrometry is insufficient to fully characterize orbits of Lazuli exoplanet targets.

Absolute astrometry is indispensable in constraining the orbit of Jovian exoplanets such that they can be efficiently observed by the Lazuli ESC. However, many of the exoplanets that will be targeted by ESC have orbital periods longer than the 66-month temporal baseline of Gaia DR4’s astrometry. Thus, these planets will lack the astrometric coverage essential to confidently

constrain their orbits and inform efficient direct imaging campaign design. E. C. Matthews et al. 2024 demonstrated the detrimental effects of poor orbital coverage on predictions of a planet’s position. In their work, they predicted the position of the exoplanet ϵ Ind Ab to be southwest of the host star at time of observation with JWST/MIRI. The observations instead revealed the planet to be on the opposite side of the star. This location disparity was later determined to result in part from incomplete orbital coverage in the RV data used in predicting the planet’s position (E. C. Matthews, 2026). Unlike in the case of JWST where this was not an issue, this sort of predicted location disparity would be catastrophic to direct imaging observations with ESC as the planet would land outside of the semicircular dark hole where it can be imaged. It is therefore crucial to the design of ESC’s exoplanet observations that planet orbits are well-constrained by radial velocities *and absolute astrometry*.

The targets selected for this program will have portions of their orbits without astrometric measurements released with Gaia DR4. This program will produce the absolute astrometry that will fill this orbital coverage gap for these targets.

This Proposal: HST/FGS sub-milliarcsecond astrometry will fill the Gaia DR4 gap

The targets selected for this program are marked as high priority for observation with Lazuli ESC: they are stars hosting known Jovian planets with orbital periods too long to be completely covered by the 66-month span of astrometric measurements available in Gaia DR4. The HST Fine Guidance Sensors have been previously established to provide valuable, precise absolute astrometry capable of constraining the orbits of Jovian exoplanets (e.g. G. F. Benedict et al. 2017, W. Thompson et al. 2025).

We propose HST/FGS observations of the selected exoplanet host stars to obtain sub-milliarcsecond precision absolute astrometry that will fill the Gaia DR4 astrometry gaps. For each target we request 3 independent visits, each separated by approximately 4 months. Each visit will consist of 2 orbits of observation with HST/FGS with individual exposures of 100 s. This observation strategy ensures 1) sufficient independent HST/FGS observations are performed during each visit to achieve sub-milliarcsecond astrometric precision, and 2) sufficient time between visits to observe significant changes in the star’s absolute position on-sky. Figure 1 demonstrates how HST/FGS data following this observing strategy has previously been used in constraining the orbit of $1M_{\text{jup}}$ exoplanet ϵ Eri b.

This program will ultimately produce a catalog of absolute astrometric measurements for these exoplanet host stars. This catalog will fill the astrometry gap left by the limited temporal coverage of Gaia DR4 and will be used in the orbit fitting needed to prepare for exoplanet direct imaging observations with Lazuli ESC.

This program enables the next generation of exoplanet direct imaging science with Lazuli.

The proposed HST/FGS astrometry for these high-priority Lazuli ESC Jovian exoplanets will enable the efficient scheduling of the direct imaging campaigns that will image these planets for the first time. With the proposed HST/FGS astrometry, the positions of these Jovian planets relative to their host stars will be accurately predicted, enabling 1) the scheduling of observations during periods of high probability of detection and 2) the placement of the Lazuli ESC dark hole appropriate for the predicted position of the planet.

The orbital coverage gaps left behind by Gaia DR4 must be resolved by this program in preparation for Lazuli’s launch around 2030 as Gaia DR5 covering the entire 12-year Gaia mission will not be released until after Lazuli’s launch. This program aligns with the Hubble

Space Telescope’s key science theme of deepening understanding of exoplanetary systems by paving the way for the next generation of direct imaging observations. Beyond scheduling of direct imaging observations, the orbital and mass constraints of these Jovian planets resulting from this program’s astrometry catalog will allow for both robust interpretation of Lazuli ESC photometry with atmospheric models (see, e.g., Figure 4, R. E. Lupu et al. 2016), and a more complete understanding of the planetary system architectures found around nearby stars. This understanding of planetary system architecture is critical to realizing the ambitious goal of detecting Earth-like exoplanets around Sun-like stars with the Habitable Worlds Observatory.

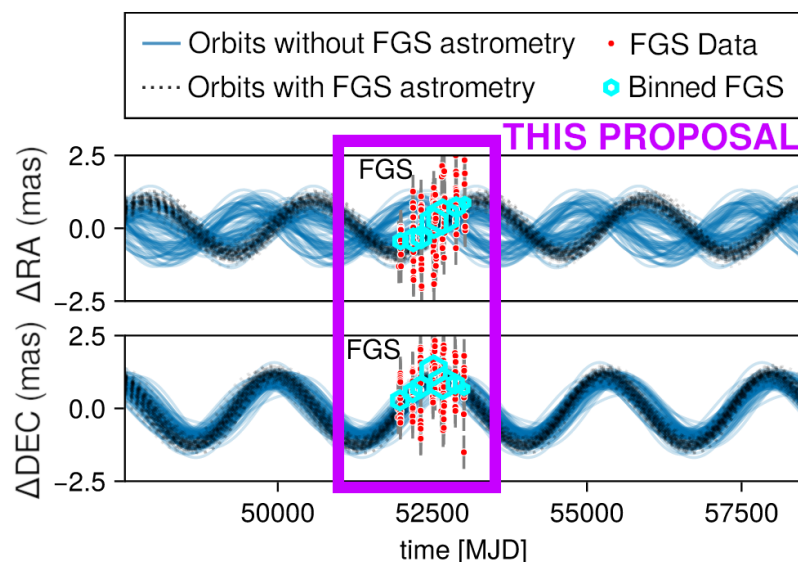


Figure 1: **Lazuli exoplanet orbit constraints will be significantly improved by this proposal’s HST/FGS astrometry.** Example shown for $1M_{\text{jup}}$ exoplanet ϵ Eri b. *Top* and *Bottom* show displacement from mean proper motion of host star ϵ Eri by in right ascension and declination respectively over time. Solid blue lines are predicted displacements of ϵ Eri resulting from ϵ Eri b orbits computed from radial velocity and astrometry sources *excluding HST/FGS data*. Dashed black lines are similarly computed *including HST/FGS data*. Individual HST/FGS astrometry measurements are shown as red points. Blue circles represent the weighted median of all individual measurements during an HST/FGS visit. Without this program’s HST/FGS data, the Lazuli ESC planets this program will observe will not have the astrometry necessary to distinguish between the two opposite orbital fitting solutions demonstrated in this plot (see the Δ RA plot, top). This program will generate the HST/FGS data necessary to similarly constrain the orbits of the selected exoplanets, enabling the efficient design of direct imaging campaigns for these planets with Lazuli ESC. Figure adapted from W. Thompson et al. 2025.

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